

HAWARDEN, United Kingdom

WMO: 033210

Lat: 53.175N

Lon: 2.987W

Elev: 45

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.5	27.4	21.4	16.1	30.0	23.9	18.1	31.9	29.8	52.0	26.4	50.3	2.3	160	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	77.5	64.8	73.7	63.3	70.8	61.9	66.5	73.9	64.8	71.6	63.1	69.2	8.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	89.1	69.6	62.0	83.4	67.7	60.5	78.8	66.2	31.1	73.6	29.8	71.7	28.6	69.3	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	20.1	17.8	DB	18.7	83.4	5.3	3.7	14.9	86.1	11.8	88.3	8.8	90.4	4.9	93.1	(4)
(5)			WB	18.6	72.4	3.9	4.3	15.8	75.5	13.5	78.0	11.3	80.4	8.4	83.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	41.0	41.7	44.1	47.6	53.0	58.2	61.4	61.1	57.0	51.7	45.7	41.7	(6)
(7)		DBStd	9.05	6.20	5.81	5.33	5.07	4.97	4.37	4.13	3.86	4.01	5.22	5.62	6.88	(7)
(8)		HDD50	1298	281	236	192	104	25	1	0	0	1	42	150	266	(8)
(9)		HDD65	5362	743	651	646	522	374	209	127	134	240	413	580	723	(9)
(10)		CDD50	1448	3	5	11	32	116	247	354	344	213	95	20	8	(10)
(11)		CDD65	36	0	0	0	0	0	6	16	12	2	0	0	0	(11)
(12)		CDH74	311	0	0	0	0	13	59	145	81	11	2	0	0	(12)
(13)		CDH80	49	0	0	0	0	0	8	24	17	0	0	0	0	(13)
(14)	Wind	WSAvg	8.1	9.2	9.4	9.0	8.1	7.9	7.7	7.2	7.2	7.1	7.7	8.0	8.4	(14)
(15)	Precipitation	PrecAvg	27.7	2.4	1.9	2.2	1.8	2.2	2.1	1.8	2.2	2.4	2.7	3.1	2.9	(15)
(16)		PrecMax	33.9	3.8	4.8	3.7	3.5	6.3	4.4	4.3	4.0	10.1	5.5	7.3	5.6	(16)
(17)		PrecMin	20.1	0.7	0.1	0.3	0.1	0.4	0.6	0.5	0.1	0.2	0.4	1.5	0.9	(17)
(18)		PrecStd	3.1	0.8	1.1	0.9	1.0	1.5	1.0	1.0	0.9	1.9	1.2	1.4	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	57.9	61.1	69.5	75.7	80.9	83.2	82.3	75.4	67.1	60.8	59.3	(19)
(20)			MCWB	52.7	53.1	52.2	56.4	62.2	65.9	66.3	66.2	64.0	60.6	56.9	53.9	(20)
(21)		2%	DB	54.6	54.8	57.6	63.3	70.3	74.2	77.9	75.1	70.7	64.0	58.4	55.9	(21)
(22)			MCWB	50.8	50.8	51.0	53.9	59.7	62.8	65.2	64.5	61.7	58.9	54.9	52.3	(22)
(23)		5%	DB	52.4	52.9	55.1	59.8	65.7	70.4	74.0	71.4	67.5	61.9	56.4	54.0	(23)
(24)			MCWB	48.8	49.0	49.7	52.1	57.0	61.0	63.9	63.1	59.9	57.3	53.0	50.8	(24)
(25)		10%	DB	50.4	50.9	53.0	56.8	62.4	67.3	70.4	68.7	64.9	60.0	54.4	51.6	(25)
(26)			MCWB	47.1	47.1	48.2	50.6	55.0	59.2	62.1	61.4	58.3	55.9	51.3	48.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.6	54.1	54.0	57.9	64.3	69.1	70.9	68.7	65.9	62.2	57.8	55.1	(27)
(28)			MCDWB	56.1	56.9	58.6	65.8	73.3	75.1	77.3	76.0	72.6	65.8	60.0	57.2	(28)
(29)		2%	WB	51.3	51.3	52.0	55.4	60.9	64.6	66.6	65.9	63.4	60.0	55.6	53.0	(29)
(30)			MCDWB	54.1	54.3	56.2	61.0	68.7	71.3	74.5	72.7	69.0	63.0	57.8	55.8	(30)
(31)		5%	WB	49.4	49.4	50.4	53.2	58.4	62.0	64.7	64.3	61.1	58.2	53.5	51.2	(31)
(32)			MCDWB	52.0	52.5	54.5	58.5	64.2	68.8	71.9	70.4	65.9	61.2	55.9	53.7	(32)
(33)		10%	WB	47.4	47.5	48.7	51.2	56.1	60.1	62.9	62.7	59.3	56.4	51.6	48.8	(33)
(34)			MCDWB	50.0	50.5	52.6	56.1	61.2	66.0	69.4	67.4	63.7	59.4	54.0	51.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	11.1	13.4	15.6	16.1	15.3	15.5	14.8	14.9	13.0	10.9	10.2	(35)
(36)			MCDBR	11.6	12.3	16.2	20.6	22.5	22.4	23.3	19.5	19.2	14.2	11.7	11.5	(36)
(37)			MCWBR	10.3	9.7	11.1	12.8	13.5	13.3	12.7	10.7	11.4	10.1	9.8	10.1	(37)
(38)		5% WB	MCDWB	11.4	11.8	15.2	18.3	20.6	19.2	20.1	17.2	16.6	13.5	11.2	11.0	(38)
(39)			MCWBR	10.4	9.8	11.3	12.4	13.3	12.7	12.6	10.5	11.0	10.6	9.9	10.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.341	0.372	0.396	0.400	0.396	0.396	0.381	0.369	0.351	0.326	0.316	(40)	
(41)		taud	2.318	2.367	2.288	2.239	2.271	2.313	2.320	2.377	2.410	2.423	2.379	2.292	(41)	
(42)		Ebn at Noon	203	240	255	264	269	271	269	267	255	234	203	181	(42)	
(43)		Edh at Noon	19	25	33	39	40	38	38	34	29	23	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	209	422	786	1214	1534	1614	1522	1248	906	504	252	159	(44)	
(45)		RadStd	20	46	72	118	105	155	154	85	68	44	27	16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page